UC IPM

‹ Corn

AGRICULTURE

Cutworms

Black cutworm: *Agrotis ipsilon*

Variegated cutworm: *Peridroma saucia*

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On this page

Description of the Pest

Adults are moths approximately 1 inch long with a wing span of 1.25 to 2 inches and vary widely in coloration. Eggs are somewhat flattened on top, white to dull or off-white in color, and ribbed. They are generally deposited in massed rows. Eggs may be deposited on crop foliage, but are frequently found on weeds. Fully grown larvae range from 1 to 1.75 inches in length and commonly curl into a **C-shape** when disturbed.

Cutworms are most active and cause the most damage during spring and early summer months. The larvae normally hide under debris on the soil surface during the day, but are active, voracious feeders at night. Some cutworms climb into the host plant to feed, but many stay on the ground, cutting seedling host plants off at or just below the soil surface.

Damage

Cutworms cut young plants off at the base or near the ground level. Usually, it is necessary to dig in the soil to find cutworm larvae and to determine the extent of the infestation and the size of the cutworms involved.

Management

If the cutworm population is reducing the plant stand, treat during the seedling stage. Frequently, the damage is most serious at the edges of a field, but stand loss can occur in a spotty pattern throughout the field. Treatment of hot spots may be possible. Seedlings will regrow if damage is above the growing point.

Organically Acceptable Methods

Eliminating weeds 2 weeks before planting both within and adjacent to the field can help to minimize cutworm problems in an organically managed crop.

Pesticides and Natural Enemy Releases

Not all registered pesticides are listed. **The following are ranked by their IPM value, with the most effective and least**

harmful to natural enemies, honey bees, and the environment listed at the top of the table. When choosing a pesticide, consider information related to water and air quality, resistance management, and the pesticide's properties and application timing. Use PestManage to compare summarized management options for different pests in the same crop. Always carefully read the label of the product being used and take all necessary precautions when handling pesticides.

Expand table

Rank	Active ingredient	Example trade name
Application Timing Varies (See *UC IPM Pest Manageme*		
A	permethrin	Ambush 25W
B	carbaryl (bait)	Sevin 5 Bait

—	No information.
I	Do not apply or allow to drift to plants that are flowering including weeds. Do not allow pesticide to contaminate water accessible to bees including puddles.
II	Do not apply or allow to drift to plants that are flowering including weeds, except when the application is made between sunset and midnight if allowed by the pesticide label and regulations. Do not allow pesticide to contaminate water accessible to bees including puddles.
III	No bee precaution, except when required by the pesticide label or regulations.

- ‡ a b Restricted entry interval (REI) is the number of hours (unless otherwise noted) from treatment until the treated area can be safely entered without personal protective equipment. Preharvest interval (PHI) is the number of days from treatment to harvest. In some cases, the REI exceeds the PHI. The longer of the two intervals is the minimum time that must elapse before harvest.
- [1] ↩ Group numbers for insecticides and miticides are assigned by the Insecticide Resistance Action Committee (IRAC). Insecticides with unknown modes of action are assigned mode-of-action group numbers (MoAs) that begin with UN. Rotate pesticides with a different mode-of-action group number, and do not use products with the same mode-of-action group number more than twice per season to help prevent the development of resistance. For example, the organophosphates have a group number of 1B; insecticides with a 1B group number should be alternated with insecticides that have a group number other than 1B.
- [2] ↩ Range of insect and mite groups affected by a pesticide. **Broad** means the pesticide affects most groups of insects and mites; **narrow** means the pesticide affects only a few specific groups.
- [3] ↩ Risk of harm to honey bees. For more information, see *Bee Precaution Pesticide Ratings*.
- [4] ↩ Risk of harm to predatory mites. Toxicities are generally to western predatory mite, *Galendromus occidentalis*. Where differences have been measured in toxicity of the pesticide-resistant strain versus the native strain, these are listed as pesticide-resistant strain or native strain.
- [5] a b Risk of harm to parasitoids and general predators. Toxicities are averages of reported effects and should be used only as a general guide. Actual toxicity of a specific insecticide depends on factors including the application rate, environmental conditions, and the life stage and species of a parasitoid or predator.

- 6 ↩ Length of time residue affects natural enemies. **Short** means hours to days; **moderate** means days to 2 weeks; and **long** means many weeks or months.
- 7 ↩ Risk of harm to fish from leaching, based on USDA Natural Resources Conservation Service Windows Pesticide Screening Tool (WIN-PST).
- 8 ↩ Risk of harm to fish from adsorbed runoff, based on USDA Natural Resources Conservation Service Windows Pesticide Screening Tool (WIN-PST).
- 9 ↩ Risk of harm to fish from solution runoff, based on USDA Natural Resources Conservation Service Windows Pesticide Screening Tool (WIN-PST).
- 10 ↩ Risk of harm to humans from leaching, based on USDA Natural Resources Conservation Service Windows Pesticide Screening Tool (WIN-PST).
- 11 ↩ Risk of harm to humans from solution runoff, based on USDA Natural Resources Conservation Service Windows Pesticide Screening Tool (WIN-PST).
- 12 ↩ Date information was last updated in the *UC IPM Pest Management Guidelines*.

UC PEER REVIEWED

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